
CommunityKV: Efficient Long-Context Decoding via Graph Partitioning

Anonymous Author(s)

Affiliation

Address

email

Abstract

Scaling Transformers to long contexts is constrained by the quadratic cost of self-attention and the linear growth of key-value cache memory transfer. Sparse attention mitigates this by retrieving only relevant tokens, but current approaches either require retraining or, within the training-free regime, rely on semantically coarse heuristics or expensive clustering that cannot be updated efficiently during decoding. We introduce CommunityKV, a framework that formulates sparse attention as a community detection problem. CommunityKV constructs a token graph from the QK^T scores already computed during standard prefill and partitions it into communities to enable community retrieval. A local update rule assigns newly generated tokens to communities in constant time, enabling sparse retrieval throughout streaming decoding without global re-partitioning. We evaluate CommunityKV on open weight models and long context benchmarks, demonstrating that our approach achieves up to $1.86\times$ higher decoding throughput than exact FlashAttention-2 with negligible accuracy degradation.

1 Introduction

The capacity for Transformers [36] to process long context inputs powers a new generation of applications, ranging from repository-level code understanding [2, 29] and long-horizon agentic planning [27, 26] to multimodal reasoning [38]. However, as the input context length grows, standard self-attention encounters a prohibitive bottleneck: the quadratic cost of attention and the linear growth of key-value (KV) cache memory transfer.

Despite the increasing KV cache size for modern applications, empirical studies demonstrate that token prediction accuracy is driven by only a small fraction of relevant past tokens [46, 13]. Sparse attention methods attempt to exploit this redundancy by loading only a portion of the KV cache at each generation step [34, 24]. Recent architectural methods such as DeepSeek Sparse Attention [9] integrate sparsity into the training objective, but require large-scale training. Within the training-free regime, current approaches force a compromise: they either employ retrieval heuristics that suffer from unacceptable accuracy degradation [33] or rely on heavy preprocessing of the KV cache that is difficult to update during decoding [16].

In this work, we propose CommunityKV, a framework that formulates sparse attention as community detection on a graph induced from attention scores. Rather than clustering key vectors or heuristically partitioning token blocks, CommunityKV leverages the structure already present in the QK^T matrix to identify semantically coherent token groups. A local update rule extends the partition to newly generated tokens in constant time, enabling streaming decoding without re-clustering.

Our contributions are as follows:

- **Methodology:** We introduce a novel training-free framework that formulates sparse attention as community detection on a token graph induced from $QK^\top$ scores. A local update rule integrates newly generated tokens into the partition in constant time, supporting streaming decoding without global re-partitioning.
- **Theoretical analysis:** We derive complexity bounds demonstrating that graph construction is amortized within standard prefill, with sub-linear retrieval and constant per-step community updates during decoding.
- **Empirical performance:** We evaluate CommunityKV on long context benchmarks, demonstrating up to $1.86\times$ higher decoding throughput at less than 8% prefill overhead, while maintaining the accuracy of exact attention. CommunityKV bridges the gap between efficiency and fidelity: it consistently outperforms heuristic-based approaches on long context comprehension tasks while bypassing the computational overhead of geometric clustering.

2 Related work

Architectural vs. training-free sparse attention Recent methods integrate sparsity directly into the training objective, achieving large throughput gains. DeepSeek Sparse Attention (DSA) [9] scores queries against compressed key representations, while Native Sparse Attention (NSA) [43] combines block-level token compression with fine-grained token selection. However, these approaches couple sparsity to model training: the upfront cost amortizes only at very high inference volumes, and the resulting sparsity pattern is fixed at training time, locking retrieval quality to the training distribution.

CommunityKV targets the complementary setting: training-free sparse attention. Within the training-free regime, existing approaches reduce KV cache traffic via two strategies. Compression methods [46, 22] enforce strict memory ceilings through token eviction or merging, but are inherently lossy: once discarded, information cannot be recovered. Selective retrieval methods instead preserve the full KV cache in memory and dynamically load only relevant tokens at each decoding step. The effectiveness of retrieval-based methods depends on the design of the sparsity mask, which typically employs either fixed or dynamic patterns.

Fixed-pattern sparse attention Early work on sparse attention relied on predefined, query-independent sparsity patterns. The Sparse Transformer [7] introduced factorized attention with strided patterns, achieving sub-quadratic $\mathcal{O}(n\sqrt{n})$ complexity. Longformer [4] and BigBird [44] combined sliding windows, dilated patterns, and global tokens to achieve linear $\mathcal{O}(n)$ complexity. LongNet [10] extended dilated attention to billion-token sequences, and StreamingLLM [41] showed that retaining initial “attention sinks” stabilizes sliding window attention. Because these patterns are fixed in advance, they miss relevant tokens that fall outside the predefined structure.

Dynamic sparse attention Dynamic sparse attention frames the problem as query-dependent retrieval, typically using auxiliary data structures to organize the KV cache and select relevant subsets at each step. A first family uses heuristics over contiguous token blocks. MoBA [23] applies block-wise gating to filter context chunks, while Quest [33] tracks channel-wise min-max values to estimate query-key overlap. These methods avoid heavy auxiliary data structures but treat blocks of contiguous tokens as the retrieval unit: a high-scoring outlier within a block triggers retrieval of the entire block, including unrelated neighbors. A second family uses Locality Sensitive Hashing [17, 45, 6, 15] to bucket tokens via random projections. Because the projections are data-independent, recall depends on the number of hash tables, and the tables required for high recall offset the per-table efficiency gain. A third family applies k-means clustering to the key vectors [30, 16, 37, 39]. These methods produce semantically coherent clusters, but iterative cluster optimization is expensive at long context, and the cluster structure does not admit a constant-time update rule for autoregressive generation.

Comparison to GraphKV GraphKV [21] is the most closely related work, constructing a sparse attention graph and applying a “decay signal propagation” algorithm to evict semantically redundant tokens. The result is a compressed cache that every subsequent decoding step queries against. CommunityKV adopts the opposite philosophy. Instead of treating semantic similarity as a signal to be decayed, we treat it as the basis for community formation: we retain the full KV cache and dynamically retrieve a different subset of coherent communities for each query, allowing the model to access locally dense semantic structure without permanently deleting context.

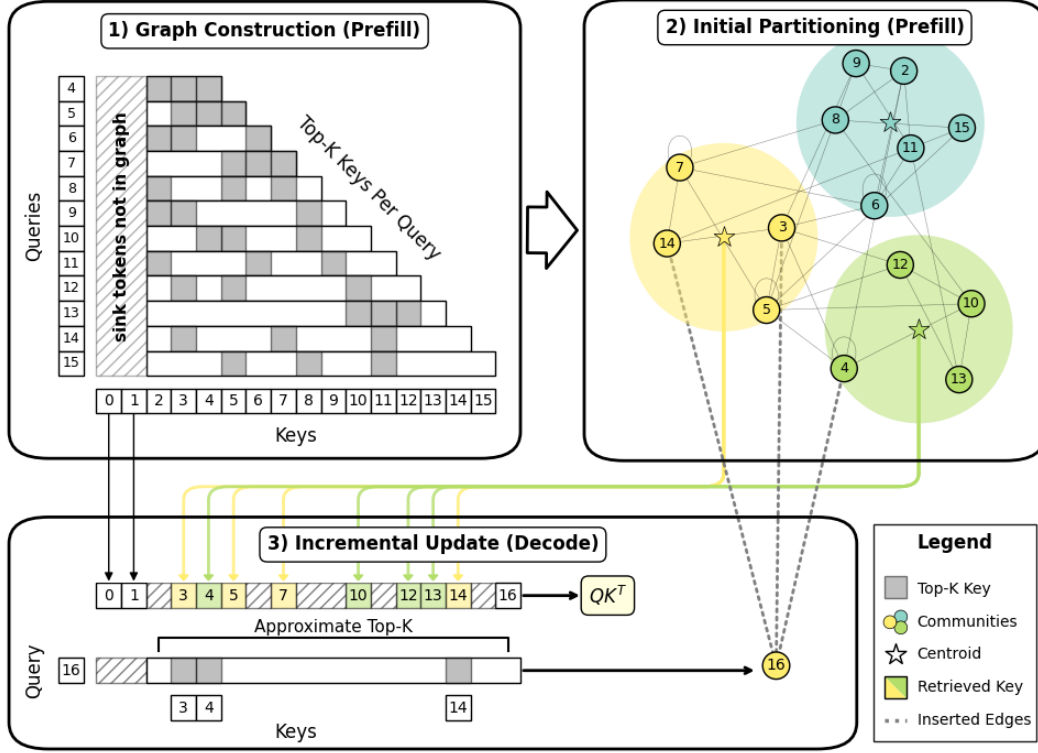


Figure 1: Overview of the CommunityKV framework. **(1) Graph Construction.** During prefill, we induce a sparse graph from the attention matrix by connecting each query to its top- κ keys (gray cells, $\kappa = 3$ shown). Sink tokens (hatched) are excluded from the top- κ pool. **(2) Initial Partitioning.** We partition the graph into semantic communities (colored regions) using the Leiden algorithm and compute a centroid (star) per community. **(3) Incremental Update.** At each decoding step (token 16 shown), we score the query against community centroids to retrieve a candidate subset, run exact attention on this subset together with sink tokens, and use the resulting top- κ keys to update the new token’s edges and assign it to a community in constant time. For clarity, only direct attention $w^{(1)}$ relations are depicted; this corresponds to $\lambda = 1$ in the combined graph.

87 3 Method

88 Our approach has three phases (Figure 1): (1) graph construction, where we build a sparse token
 89 graph from attention scores during prefill; (2) initial partitioning, where we cluster the graph into
 90 semantic communities; and (3) incremental update, where newly generated tokens are assigned to
 91 communities in constant time during decoding.

92 3.1 Graph construction

93 We construct a graph where nodes represent tokens and edges encode two kinds of relations: direct
 94 attention, where one token attends strongly to another, and co-attention, where two tokens are jointly
 95 attended by a common query. Let $\alpha \in [0, 1]^{n \times n}$ be the attention score matrix over an n -token context,
 96 where α_{ij} is the score of query i attending to key j , and let $\mathcal{K}_i$ be the top- κ key positions for query i :

$$\mathcal{K}_i = \arg \operatorname{top-}\kappa_{j=|\mathcal{S}|+1, \dots, i} \alpha_{ij}. \quad (1)$$

97 We exclude the set of initial “sink token” positions $\mathcal{S}$, as they exhibit disproportionately high attention
 98 scores regardless of semantic relevance [40].

99 **Direct attention graph** The direct attention graph captures explicit query-key dependencies: a
 100 directed edge $i \rightarrow j$ records that query i attends strongly to key j :

$$w_{ij}^{(1)} = \alpha_{ij} \mathbf{1}(j \in \mathcal{K}_i) \quad (2)$$

101 where $\mathbf{1}(\cdot)$ is the indicator function.

102 **Co-attention graph** If two keys are strongly attended by the same query, they play similar roles in
 103 that query’s context. The co-attention graph captures this by linking keys that co-occur in the top- κ
 104 set of a common query:

$$w_{ij}^{(2)} = \sum_{m=|\mathcal{S}|+1}^n \alpha_{mi} \alpha_{mj} \mathbf{1}(\{i, j\} \subseteq \mathcal{K}_m). \quad (3)$$

105 **Combined graph** We combine the two adjacency matrices into a single symmetric adjacency
 106 matrix:

$$w = \frac{\lambda}{2} [w^{(1)} + (w^{(1)})^\top] + (1 - \lambda) w^{(2)}, \quad (4)$$

107 with $0 \leq \lambda \leq 1$ controlling the balance between direct and co-attention relations. By construction,
 108 the graph is sparse: each of the n queries contributes κ edges to $w^{(1)}$ and at most κ^2 edges to $w^{(2)}$,
 109 so the total edge count satisfies $E = \mathcal{O}(n\kappa(1 + \kappa)) = \mathcal{O}(n)$ for constant κ .

110 The combined graph admits a dynamical interpretation that motivates graph partitioning. Since α
 111 is row-stochastic, it defines a transition kernel for a discrete Markov chain over the token sequence.
 112 Symmetrizing as $(\alpha + \alpha^\top)/2$ and augmenting with the symmetric second-order structural equivalence
 113 relations $\alpha^\top \alpha$ induces a time-reversible random walk. In this framework, token communities
 114 correspond to metastable states of the walk: groups of tokens the walk is unlikely to escape over time.
 115 We partition the tokens into these communities by maximizing graph modularity [25], a standard
 116 objective that is formally equivalent to maximizing Markov Stability [20]. In practice, we apply the
 117 Leiden algorithm to the sparsified kernel w .

118 3.2 Initial partitioning

119 **Leiden community detection** The Leiden algorithm [35] maximizes modularity:

$$Q = \frac{1}{2m} \sum_{ij} \left(w_{ij} - \gamma \frac{w_i w_j}{2m} \right) \mathbf{1}(c_i = c_j), \quad (5)$$

120 where $m = \frac{1}{2} \sum_{ij} w_{ij}$ is the total graph weight, $w_i = \sum_j w_{ij}$ is the weighted degree of token i , and
 121 c_i is the community label of token i . The resolution parameter γ controls community granularity:
 122 larger values favor smaller, denser communities. Leiden iterates three phases: fast local moving,
 123 refinement, and aggregation, improving on Louvain [5] by guaranteeing γ -connected communities
 124 and converging to subset-optimal partitions [35].

125 We choose Leiden over alternatives for several reasons. Leiden empirically exhibits $\mathcal{O}(E)$ run-
 126 time [35] and few iterations required for convergence; [5] report fewer than 5 across all tested
 127 networks for the analogous Louvain algorithm. We cap iterations at $r = \lfloor \log_{10} n \rfloor$, yielding $r \in [4, 7]$
 128 for our evaluation contexts (10k–10M tokens). With graph sparsity bounded at $E = \mathcal{O}(n)$, parti-
 129 tioning costs $\mathcal{O}(rE) = \mathcal{O}(n \log n)$. We then compute community centroids in a single $\mathcal{O}(nd)$ pass,
 130 yielding a clustering cost of $\mathcal{O}(n \log n + nd)$ (Table 1). By contrast, Multipole Attention assigns n
 131 tokens to $\mathcal{O}(n)$ centroids via k-means, yielding $\mathcal{O}(n^2 d)$ clustering cost. Because pairwise similarity
 132 is already captured in the graph edges via $QK^\top$, our clustering operates on scalar weights rather than
 133 d -dimensional vectors.

134 Leiden’s local-moving phase is embarrassingly parallel; each node’s modularity-gain computation
 135 ΔQ (Eq. 7) depends only on its neighbors and community-level aggregates, so all nodes can be evalu-
 136 ated simultaneously. Recent parallel implementations exploit this structure to achieve a processing
 137 rate of 403M edges/s on a 3.8B-edge graph using 64 CPU threads [32], and NVIDIA’s cuGraph [11]
 138 reports significant speedup over CPU baselines.

139 Leiden admits incremental update schemes that avoid global re-partitioning when edges are added [31,
 140 14]. Our local update rule (Section 3.3) is a domain-specific specialization of this idea, exploiting the
 141 structure of autoregressive decoding to assign new tokens in $\mathcal{O}(d)$ time per step.

Table 1: Computational complexity comparison. n is sequence length, d is hidden dimension. Note that our method achieves sub-linear retrieval $\mathcal{O}(\sqrt{nd})$ without expensive clustering or update costs.

Method	Initialization Overhead	Per-step Update	Per-Step Retrieval
Quest	$\mathcal{O}(nd)$	$\mathcal{O}(d)$	$\mathcal{O}(nd)$
Multipole	$\mathcal{O}(n^2d)$	$\mathcal{O}(nd)$	$\mathcal{O}(nd)$
CommunityKV	$\mathcal{O}(n \log n + nd)$	$\mathcal{O}(d)$	$\mathcal{O}(\sqrt{n}d)$

3.3 Incremental update

Community retrieval During decoding, we retrieve relevant tokens by scoring the latest query q_i against community centroids and aggregating positions from the top-ranking communities into a candidate set $\mathcal{R}_i$. We then form the sparse context $\mathcal{S} \cup \mathcal{R}_i \cup \{i\}$ by augmenting $\mathcal{R}_i$ with the attention sinks $\mathcal{S}$ and the current query position i , truncating to a fixed memory budget. Because the number of communities scales as $\mathcal{O}(\sqrt{n})$ [12] (Figure 2, Appendix A), the cost of retrieval is $\mathcal{O}(\sqrt{n}d)$ (Table 1).

Edge approximation To integrate the new token into the graph, we approximate its edge weights using only the retrieved set. We compute approximate attention scores and top- κ keys over $\mathcal{R}_i \cup \{i\}$:

$$\begin{aligned}\tilde{\alpha}_{ij} &= \text{softmax}(q_i^\top k_j)_{j \in \mathcal{R}_i \cup \{i\}}, \\ \tilde{\mathcal{K}}_i &= \arg \text{top-}\kappa \tilde{\alpha}_{ij}. \\ &\quad j \in \mathcal{R}_i \cup \{i\}\end{aligned}\tag{6}$$

Substituting these approximations into Eqs. (2) and (3) yields the edge updates. We add κ direct attention edges $w_{ij}^{(1)} = \tilde{\alpha}_{ij}$ for each $j \in \tilde{\mathcal{K}}_i$, and increment the κ^2 co-attention edges between pairs $j, k \in \tilde{\mathcal{K}}_i$ by $\Delta w_{jk}^{(2)} = \tilde{\alpha}_{ij} \tilde{\alpha}_{ik}$.

Greedy community assignment After integrating the new token into the graph, we assign it a community label. We freeze the existing community assignments and consider two options: form a new singleton community $\{i\}$ or join a neighboring community $C \in \{c_j \mid j \in \tilde{\mathcal{K}}_i\}$. We pick the option that maximizes the modularity gain:

$$\Delta Q(i \rightarrow C) \propto \sum_{j \in C \cap \tilde{\mathcal{K}}_i} w_{ij} - \gamma \frac{w_i w_C}{2m},\tag{7}$$

where $w_C = \sum_{j \in C} w_j$ is the total weight of community C .

Each decoding step touches at most $\kappa(\kappa + 1)$ edges, evaluates κ candidate communities, and updates one centroid in $\mathcal{O}(d)$. Since κ is a small constant, the total per-step update cost is $\mathcal{O}(d)$, independent of sequence length n (Table 1).

Stability of incremental updates Our local update rule maintains partition quality throughout decoding without global re-partitioning. In a stress test over 4096 generated tokens, modularity remains in $[0.44, 0.50]$ and attention mass recall stays above 85% at a 4096-token budget (Figure 3; see Appendix B for the full experimental protocol). The KV cache and the routing graph are decoupled in CommunityKV: updates modify only the graph, leaving past KVs untouched. A single global re-partitioning restores optimal modularity in the routing structure. Running Leiden every $\mathcal{O}(\sqrt{n})$ steps adds amortized $\mathcal{O}(\sqrt{n} \log n)$ cost per step, making periodic refresh a low-cost knob for unbounded generation.

3.4 Implementation details

We fuse top- κ selection into the FlashAttention kernel: as each $b_q \times b_k$ tile is computed, a register-level bitonic sort identifies b_q sets of local top- κ candidates, which are reduced per-query via radix select. The total cost is $\mathcal{O}(n^2)$, matching the underlying prefill matrix multiplication, because both the block-level sorting and the global reduction scale quadratically in n with constant factors determined

by the tile dimensions b_q , b_k and κ (Appendix C.1). Empirically, this adds 6–8% to prefill wall time (Table 5).

Leiden partitioning is pipelined with the forward pass: as soon as layer L ’s attention scores are available, its graphs are dispatched to asynchronous workers while layers $L+1, \dots$ continue executing (Figure 4). The only exposed latency is the final layer’s partition (0.37–0.45s across the Qwen3 family). At 130k tokens with $\kappa = 8$, each per-head graph contains ~ 1 M edges and occupies < 20 MB in bfloat16; the aggregate footprint across all heads is < 1 GB under query-group aggregation. We provide the full complexity derivation, pipelining schedule, and memory accounting in Appendix C.

4 Experiments

We evaluate CommunityKV along three axes: (1) accuracy compared to sparse attention baselines on long-context reasoning benchmarks, (2) wall-clock latency on an NVIDIA H100 80GB, and (3) ablation studies over key design parameters. Unless otherwise noted, we fix $\kappa = 8$, $\lambda = 0.5$, use a 4096-token retrieval budget, exclude the first 12 sink tokens from graph construction, and maintain one independent graph per query head. All models use greedy decoding and YaRN RoPE scaling [28] to extend the model context.

4.1 Experimental setup

4.1.1 Datasets

LongBench v2 We evaluate on LongBench v2 [1], a suite of challenging multiple-choice questions spanning code understanding, long-dialogue history, and multi-document QA. Human experts achieve only 53.7% accuracy, making it a rigorous test of reasoning over long contexts rather than simple retrieval.

BABILong We additionally evaluate on BABILong [19] QA1–QA5, a multi-hop reasoning benchmark requiring retrieval of 1–5 supporting facts from arbitrarily long inputs.

4.1.2 Models

Qwen3 Model Collection We conduct our evaluations using the Qwen3 family [42], specifically the 4B, 8B, and 14B parameter variants. This allows us to assess the robustness of our sparse attention framework across diverse architectural configurations, verifying that our retrieval mechanism scales effectively independent of model capacity.

4.1.3 Baselines

FlashAttention-2 (Dense) Exact FlashAttention-2 [8] serves as the lossless reference for both accuracy and wall-clock latency.

GraphKV As the most closely related graph-based method, GraphKV [21] constructs a sparse attention graph and permanently evicts low-scoring tokens during prefill. We include it to isolate the benefit of dynamic retrieval over static eviction.

We additionally compare against two training-free retrieval methods that span the efficiency-fidelity spectrum:

Quest Quest [33] partitions consecutive tokens into fixed blocks and tracks channel-wise min-max statistics for retrieval, representing the fast but semantically coarse end of the spectrum.

Multipole Attention Multipole [16] clusters keys via k-means, representing the high-fidelity end. It serves as a performance target that our lighter approach seeks to match without the extra compute cost.

4.2 Long context understanding

Graph-based retrieval vs. KV eviction Both CommunityKV and GraphKV construct sparse graphs from attention scores, but differ in how they use the graph: GraphKV evicts tokens permanently, while

Table 2: Accuracy on LongBench v2 across token budgets (4k–32k) for GraphKV and CommunityKV.

<i>Qwen3-8B</i> (Dense: 31.4)				
	4096	8192	16384	32768
GraphKV	3.4	10.5	18.5	22.5
CommunityKV	31.1	31.2	31.4	31.4

Table 3: Accuracy on LongBench v2 across token budgets (128–4096) at cluster size 16 for Qwen3 models (4B, 8B, 14B)

	128	256	512	1024	2048	4096
<i>Qwen3-4B</i> (Dense: 27.4)						
Quest	0.2	0.8	3.0	6.6	12.5	14.1
Multipole	25.8	26.2	25.6	25.8	25.6	25.8
CommunityKV	25.4	26.1	25.8	26.6	26.2	27.4
<i>Qwen3-8B</i> (Dense: 31.4)						
Quest	5.6	9.1	15.1	17.5	22.5	24.7
Multipole	29.4	30.0	28.4	29.8	28.2	28.8
CommunityKV	28.5	28.8	29.5	30.1	29.9	31.1
<i>Qwen3-14B</i> (Dense: 32.0)						
Quest	9.3	18.1	21.5	24.7	29.0	30.6
Multipole	30.4	29.2	29.2	28.0	28.8	28.8
CommunityKV	30.2	30.0	30.5	30.7	29.7	32.0

CommunityKV retrieves dynamically. To isolate this distinction, we compare the two on LongBench v2 with Qwen3-8B, sweeping GraphKV’s budget from 4k to 32k tokens (Table 2). GraphKV requires an $8\times$ larger budget (32k tokens) to reach 22.5% accuracy, while CommunityKV achieves 31.1% with only 4k tokens, matching the dense baseline. This confirms that semantic structure should guide dynamic retrieval rather than permanent context deletion.

Comparison between training-free retrieval methods We compare CommunityKV against Quest and Multipole on LongBench v2 across the Qwen3 family (4B, 8B, 14B), sweeping token budgets from 128 to 4096 at cluster size 16. To ensure a fair comparison, we tune the resolution parameter γ so that the number of communities matches each baseline’s cluster count (see Appendix A for the γ -community-size mapping). Results are reported in Table 3; a full sweep over cluster sizes 16, 32, 64, and 128 is provided in Appendix D.

CommunityKV achieves parity with exact FlashAttention-2 across all model scales at a 4096-token budget; for example, 32.0% on Qwen3-14B, matching the dense upper bound exactly. Multipole performs strongly at restricted budgets (128–256 tokens) but plateaus as the budget increases, suggesting that k-means fails to capture the semantic structure required for complex reasoning. Quest scales with budget but trails dense by over 6% even at 4k tokens. CommunityKV bridges this gap: its accuracy rises steadily with budget, translating additional retrieval capacity directly into better accuracy.

Multi-Hop retrieval To test whether community structure aids multi-hop retrieval, we evaluate on BABILong QA1–QA5 with Qwen3-8B at 64k context, using a 4096-token budget for all retrieval-based methods (Table 4). CommunityKV achieves the highest average accuracy among sparse methods and matches or exceeds the dense baseline on QA2, QA3, and QA5: tasks requiring chaining two or more supporting facts.

4.3 Computational efficiency

To measure the latency trade-off, we benchmark prefill and per-step decoding time for CommunityKV against dense FlashAttention-2 on the Qwen3 family (4B, 8B, 14B) at the maximum native 131k context length of Qwen3 models (Table 5).

Table 4: Accuracy on BABILong QA1–QA5 (Qwen3-8B, 64k context, 4096-token budget).

	QA1	QA2	QA3	QA4	QA5	Avg
Dense	0.66	0.38	0.23	0.55	0.62	0.488
Quest	0.38	0.35	0.22	0.49	0.59	0.406
Multipole	0.50	0.26	0.14	0.53	0.60	0.406
CommunityKV	0.40	0.38	0.25	0.54	0.63	0.440

Table 5: Latency comparison between FlashAttention-2 (Dense) and CommunityKV on an NVIDIA H100 80GB at 131k context. Left: total prefill wall time with breakdown into fused top- κ selection and exposed Leiden partitioning latency. Right: average per-step decoding wall time (128 generated tokens) with breakdown into token retrieval and sparse attention. Only attention-related kernels are reported; the remaining per-step time is MLP and layer normalization.

Prefill – Total Wall Time (s)				Decoding – Avg Per-Step Wall Time (ms)			
	4B	8B	14B		4B	8B	14B
Dense	16.5	18.2	26.3	Dense	63	64	71
CommunityKV	17.8	19.4	28.2	CommunityKV	37	34	51
Top- κ Sel.	0.94	1.09	1.49	Retrieval	5.3	4.7	7.2
Leiden	0.37	0.45	0.42	Attention	6.1	5.5	8.2
<i>Overhead</i>	+7.8%	+6.6%	+7.2%	<i>Speedup</i>	1.69×	1.86×	1.39×

CommunityKV achieves up to $1.86\times$ decoding speedup (8B model) at less than 8% prefill overhead. The overhead is dominated by fused top- κ selection (0.9–1.5s), most of which is masked by the IO cost of the 130k-token prefill; the exposed Leiden partitioning adds only 0.37–0.45s. On the decode side, CommunityKV per-step latency is governed by the fixed retrieval budget rather than sequence length, yielding near-constant cost throughout the 130k-token window.

4.4 Ablations

We ablate the key design parameters of CommunityKV on LongBench v2 with Qwen3-14B at a 4096-token budget, starting from the default configuration ($\kappa = 8$, $\lambda = 0.5$, graph per query head, 12 sink tokens excluded) (Table 6).

Graph sparsity (κ) Accuracy peaks at $\kappa = 8$ (32.0%) and drops sharply at higher values (27.8% at $\kappa = 32$), indicating that excessive connectivity obscures community boundaries. Too few edges ($\kappa = 2$, 30.3%) fragments the graph.

Direct vs. co-attention (λ) Accuracy peaks at $\lambda = 0.5$ (32.0%) and degrades toward pure direct attention ($\lambda = 1$, 28.5%) or pure co-attention ($\lambda = 0$, 28.8%), confirming that neither signal is sufficient alone.

Centroid budget By default, retrieval greedily fills the token budget with tokens from top-ranked communities. We test reserving part of the budget for additional community centroids as compressed summaries ($b_c \in \{0, 4, 16, 64, 256\}$). The greedy policy ($b_c = 0$, 32.0%) consistently outperforms centroid summaries (e.g., 28.6% at $b_c = 256$): full-token depth is more valuable than compressed breadth.

Graph aggregation In group query attention (GQA) architectures, query heads share KV heads. We test three granularities for graph construction: per-query-head (default), query-group (one graph per KV group), and layer-wise (one graph per layer). Per-head (32.0%) and query-group (31.8%) perform comparably, while layer-wise collapses to 21.5%. Query-group aggregation is a viable memory-saving alternative.

Table 6: Ablation studies on LongBench v2 (Qwen3-14B, Token Budget=4096). The default configuration used in the main results is marked with gray shading .

Graph Sparsity (κ)	$\kappa = 2$ 30.3	$\kappa = 4$ 29.4	$\kappa = 8$ 32.0	$\kappa = 16$ 28.7	$\kappa = 32$ 27.8
Graph Weight (λ)	$\lambda = 0.0$ 28.8	$\lambda = 0.25$ 29.0	$\lambda = 0.50$ 32.0	$\lambda = 0.75$ 29.0	$\lambda = 1.0$ 28.5
Centroid Budget	0 32.0	4 27.5	16 29.7	64 28.4	256 28.6
Graph Aggregation	No Agg. 32.0	Query Group 31.8	Layer-Wise 21.5	– –	– –
Sink Tokens	Excluded 32.0	In-Graph 31.3	– –	– –	– –

Sink tokens Excluding sink tokens from the graph and always retrieving them (32.0%) outperforms including them in the partition (31.3%). Sinks attract attention from nearly all positions, creating high-degree hubs that bridge unrelated communities.

5 Conclusion

We presented CommunityKV, a training-free framework that formulates sparse attention as dynamic community detection on a token graph induced from $QK^\top$ scores. By partitioning the KV cache into semantic communities during prefill and retrieving relevant subsets per query during decoding, CommunityKV achieves up to $1.86\times$ decoding speedup over FlashAttention-2 while matching its accuracy on LongBench v2 and BABILong. The constant-time local update rule enables this precision to extend to streaming generation without global re-partitioning.

Limitations and future work Our greedy community assignment freezes prior assignments, which may cause partition drift over very long generation horizons; periodic re-partitioning is a practical mitigation but adds latency. The $\mathcal{O}(\sqrt{n})$ community scaling assumption relies on a general resolution-limit result and may not hold for all attention distributions. Finally, while CommunityKV is training-free, jointly fine-tuning models to produce high-modularity attention patterns is a promising direction for further gains.

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A Community scaling and resolution parameter

We empirically verify two properties of the graph partition on LongBench v2 using Qwen3-14B (Figure 2).

A.1 Resolution parameter γ

The left panel shows the relationship between the Leiden resolution parameter γ and the average community size. Increasing γ produces smaller, denser communities, providing a direct control over retrieval granularity. In our experiments, we tune γ to match the cluster count of each baseline for fair comparison (Section 4.2).

A.2 Sub-linear community scaling

The right panel shows the number of communities as a function of sequence length. The empirical scaling closely follows $\mathcal{O}(\sqrt{n})$, consistent with the resolution limit of modularity optimization [12]. At 100k tokens, the graph partitions into approximately 50 communities. This sub-linear growth is what enables $\mathcal{O}(\sqrt{n}d)$ retrieval cost: scoring a query against $\mathcal{O}(\sqrt{n})$ centroids is substantially cheaper than the $\mathcal{O}(nd)$ cost of scoring against all tokens.

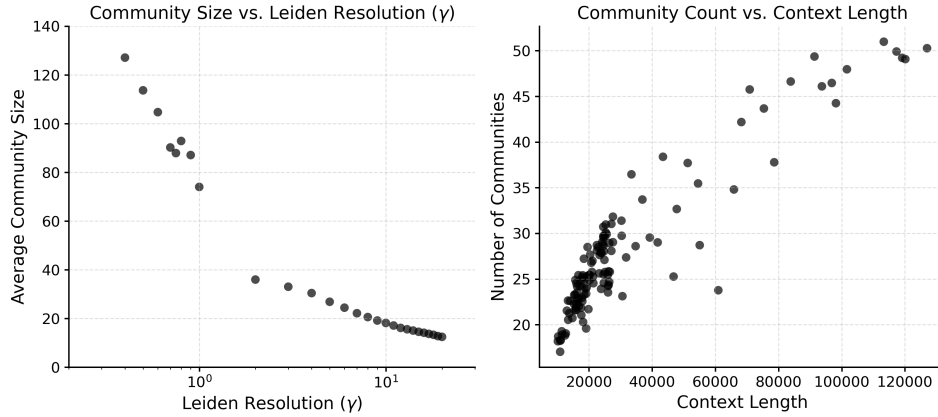


Figure 2: Empirical analysis of graph partition properties on LongBench v2 (Qwen3-14B). **Left:** Average community size as a function of the resolution parameter γ . **Right:** Number of communities vs. context length, confirming $\mathcal{O}(\sqrt{n})$ sub-linear scaling.

B Partition drift analysis

We stress-test the stability of our incremental update rule by tracking two metrics over a 4096-token generation horizon on LongBench v2 with Qwen3-14B, using the default configuration ($\kappa = 8$, $\lambda = 0.5$, 4096-token retrieval budget). No global re-partitioning is performed during generation.

B.1 Metrics

We monitor (1) graph modularity Q (Eq. 5), which measures how well the current partition reflects the evolving graph structure, and (2) attention mass recall $\tilde{\mu}_\kappa = \sum_{j \in \mathcal{R}_i} \alpha_{ij}$, the fraction of the full attention distribution captured by the retrieved set at each decoding step.

B.2 Results

Figure 3 shows both metrics as a function of generated tokens. Modularity remains in $[0.44, 0.50]$ throughout the 4096-token horizon, indicating that the greedy assignment rule continues to improve partition quality as new tokens reinforce existing community structure. Attention mass recall decays gracefully: at the 4096-token retrieval budget, recall remains above 85% after 4096 generated

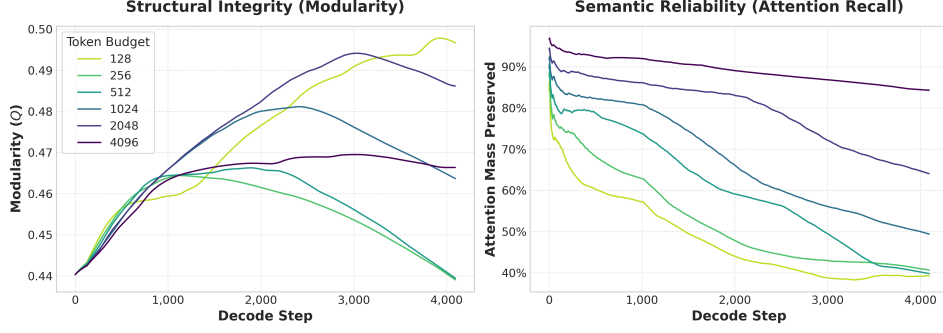


Figure 3: Partition stability over a 4096-token generation horizon (Qwen3-14B, LongBench v2). **Left:** Graph modularity Q remains stable and increases slightly as new tokens reinforce community structure. **Right:** Attention mass recall $\tilde{\mu}_\kappa$ at varying retrieval budgets. At the default 4096-token budget, recall stays above 85% throughout generation.

440 tokens. Smaller budgets (128–512 tokens) exhibit faster decay, as expected, but do not collapse
 441 catastrophically.

442 C Implementation details

443 This appendix provides the full complexity derivation for fused top- κ selection, the pipelining
 444 schedule that masks Leiden latency, and the memory accounting for per-head graphs.

445 C.1 Fused top- κ selection

446 We construct the initial sparse graph by identifying the top- κ keys for all n queries simultaneously,
 447 fusing this selection directly into the FlashAttention kernel. As the kernel computes scores for each
 448 $b_q \times b_k$ tile of queries and keys, we apply a register-level bitonic sort [3] to identify local candidates.
 449 Block-level candidates are accumulated in query-specific buffers and reduced after processing the
 450 full history. The total complexity aggregates two costs across the causal mask:

- 451 1. **Block-level sorting.** We execute a bitonic sort for every tile in the causal mask. Since
 452 there are approximately $\frac{n}{b_q} \cdot \frac{n}{b_k} / 2$ total tiles, the aggregate block-level sorting complexity is
 453 $\mathcal{O}\left(\frac{n^2}{b_q b_k} \cdot b_k \log^2 b_k\right) = \mathcal{O}\left(\frac{n^2}{b_q} \log^2 b_k\right)$.
- 454 2. **Global top- κ reduction.** For query q_i , the candidate buffer accumulates approximately
 455 $\frac{i}{b_k} \kappa$ entries, assuming i tokens are partitioned into $\frac{i}{b_k}$ causal tiles that each contribute κ
 456 candidates. We employ radix select [18] to reduce this buffer and isolate the final sparse set
 457 $\mathcal{K}_i$. Summing this linear-time operation over n queries yields $\sum_{i=1}^n \mathcal{O}\left(\frac{i}{b_k} \kappa\right) = \mathcal{O}\left(\frac{n^2}{b_k} \kappa\right)$.

458 Combining these terms, the total graph construction complexity is:

$$\mathcal{O}\left(n^2 \left(\frac{\log^2 b_k}{b_q} + \frac{\kappa}{b_k}\right)\right). \quad (8)$$

459 Because the bracketed term depends solely on the fixed constants b_q , b_k , and κ , top- κ selection scales
 460 as $\mathcal{O}(n^2)$, matching the underlying matmul and ensuring that graph construction incurs no asymptotic
 461 overhead over standard prefill.

462 The empirical overhead aligns with this bound. With $b_k = 64$ or 128 and $\kappa = 8$, the theoretical
 463 overhead ratio κ/b_k falls between 6.25% and 12.5%. Wall-clock measurements on 130k-token
 464 inputs show prefill overhead of +7.8% (Qwen3-4B), +6.6% (Qwen3-8B), and +7.2% (Qwen3-14B),
 465 confirming the kernel adds only marginal cost over standard prefill (Table 5).

466 C.2 Pipelined graph partitioning

467 Naively, Leiden partitioning would extend the prefill critical path by the sum of per-layer partition
 468 costs. We instead dispatch each layer’s partitioning to asynchronous workers immediately after its
 469 attention scores are computed, allowing it to run in parallel with the FFN of the same layer and the
 470 attention of subsequent layers (Figure 4). Because the FFN and following layers’ attention together
 471 exceed the per-layer Leiden cost, all layers except the last have their partition cost fully masked.

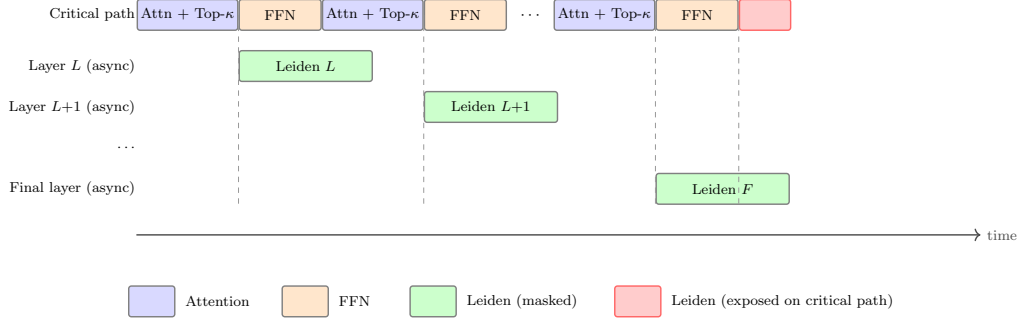


Figure 4: Pipeline schedule for a single decoding step. Leiden partitioning for each layer is dispatched asynchronously after its attention scores are computed, overlapping with subsequent layer execution. The red segment indicates the residual Leiden latency that remains exposed on the critical path when the final layer’s partitioning has not completed by the time it is needed.

472 The exposed latency is therefore the cost of partitioning a single layer’s graphs, which we measure at
 473 0.37–0.45 s across the Qwen3 family on an NVIDIA H100 (Table 5). This corresponds to 2–3% of
 474 total prefill wall time on a 130k-token input.

475 D Comprehensive evaluation and cluster size ablation

476 Table 7 reports the full grid of results across cluster sizes 16, 32, 64, and 128 for all three methods
 477 and all three model scales. The main text (Table 3) reports only cluster size 16; here we show that
 478 CommunityKV at cluster size 16 consistently dominates across budgets, while larger cluster sizes
 479 degrade performance for both CommunityKV and Quest. Multipole is relatively insensitive to cluster
 480 size, consistent with k-means producing geometrically uniform clusters regardless of granularity.

Table 7: Average accuracy on LongBench v2 for Qwen3 models across varying cluster sizes (16, 32, 64, 128) and token budgets (128, 256, 512, 1024, 2048, 4096). The FlashAttention-2 (Dense) baseline performance is provided in parentheses for each model size. **Bold** values indicate the highest accuracy for each model and token budget.

	Quest						Multipole						CommunityKV					
	128	256	512	1K	2K	4K	128	256	512	1K	2K	4K	128	256	512	1K	2K	4K
<i>Qwen3-4B</i> (Dense: 27.4)																		
16	0.2	0.8	3.0	6.6	12.5	14.1	25.8	26.2	25.6	25.8	25.6	25.8	25.4	26.1	25.8	26.6	26.2	27.4
32	0.0	0.6	1.8	5.0	9.5	12.1	26.4	26.6	26.6	26.2	26.8	26.2	21.9	22.1	24.1	25.0	23.4	25.6
64	0.0	0.2	1.0	4.6	7.8	10.9	26.0	27.0	27.0	27.4	27.4	26.0	13.9	21.6	22.6	22.2	21.6	18.2
128	0.0	0.0	0.6	2.2	4.8	5.0	27.2	27.4	26.9	25.2	25.8	27.4	9.7	14.6	12.3	13.8	16.5	17.7
<i>Qwen3-8B</i> (Dense: 31.4)																		
16	5.6	9.1	15.1	17.5	22.5	24.7	29.4	30.0	28.4	29.8	28.2	28.8	28.5	28.8	29.5	30.1	29.9	31.1
32	1.2	3.6	7.6	12.3	19.5	20.7	29.4	29.0	28.0	28.0	28.0	29.0	17.7	20.9	21.1	22.1	23.4	23.2
64	0.4	1.8	3.4	10.7	14.1	18.9	28.4	28.6	27.8	28.4	29.0	29.2	9.5	25.2	24.8	25.2	27.3	17.3
128	0.2	0.6	0.8	2.0	6.4	12.3	28.8	26.6	26.8	27.6	28.6	27.2	8.0	8.8	10.9	14.3	20.4	21.8
<i>Qwen3-14B</i> (Dense: 32.0)																		
16	9.3	18.1	21.5	24.7	29.0	30.6	30.4	29.2	29.2	28.0	28.8	28.8	30.2	30.0	30.5	30.7	29.7	32.0
32	6.6	11.5	17.3	24.1	25.0	27.4	29.2	30.4	29.0	28.4	27.8	29.4	23.4	24.8	25.0	25.2	24.7	28.0
64	4.2	7.4	11.1	17.3	20.1	23.9	30.6	30.0	29.2	29.4	27.4	28.8	14.0	25.9	26.4	27.0	27.0	23.6
128	1.2	4.0	6.2	9.1	13.9	18.7	31.0	30.2	29.8	30.2	28.8	29.0	15.7	20.5	19.0	21.1	17.6	21.5

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